

Discrete d'Alembertian and Wave Propagation on a Causal Set: A First Numerical Study

Shivangi Gupta
github.com/pikaism/Causal-set-wave

1. Motivation and Goal

This project implements a first working numerical toolkit for exploring causal set phenomenology in 1+1D Minkowski spacetime, built as independent preparation for research in the causal set approach to quantum gravity. The goal was to implement, test, and honestly validate the core kinematical and dynamical ingredients used in this field: Poisson sprinkling, the causal order, the discrete Benincasa-Dowker (BD) d'Alembertian, and a scalar field evolution scheme, compared against the known continuum limit. This toolkit is intended as a foundation for more advanced numerical studies of kinematical and dynamical aspects of causal sets, in line with ongoing work in the Theoretical Physics Group at RRI.

2. Methods

2.1 Sprinkling and causal order

N points are sprinkled via a Poisson process (uniform in spacetime volume) into a bounded region $[0, T] \times [-L, L]$ of 1+1D Minkowski spacetime. This construction is Lorentz-invariant in distribution. The causal order between any two elements is determined by the Minkowski interval condition:

$$(t_j - t_i)^2 - (x_j - x_i)^2 \geq 0 \text{ and } t_j > t_i$$

2.2 Layers and the discrete d'Alembertian

For each element x , its causally preceding elements are classified into layers L_0, L_1, L_2 according to the cardinality of the causal interval between them (the number of elements causally between the two). In 2D (1+1), the Benincasa-Dowker operator takes the form:

$$(B\phi)(x) = (1/l^2) [-2\phi(x) + 4\Sigma L_0 - 8\Sigma L_1 + 4\Sigma L_2]$$

where $l^2 = 1/\rho$ and $\rho = N / \text{Vol}(R)$ is the sprinkling density over the sprinkled region R .

Validation. A naive single-sprinkling test of this operator on simple fields (constant, linear) gave results with large, seemingly inconsistent deviations from the expected value of zero. This was diagnosed, not as an implementation bug, but as a genuine, literature-documented feature: the BD operator's convergence to the continuum d'Alembertian is a statement about the *ensemble average* over many sprinklings, not any single realization [2,3,4]. This was confirmed via ensemble averaging over 30 independent sprinklings ($N=500$): mean = 0.117, standard error = 1.365, consistent with the theoretical value of 0.

2.3 Wave packet initialization

A Gaussian wave packet is initialized on an early time-slice of the causal set:

$$\begin{aligned}\phi(t_0, x) &= A \cdot \exp(-(x-x_0)^2 / 2\sigma^2) \cdot \cos(k_0(x-x_0)) \\ \partial\phi/\partial t(t_0, x) &= -v \cdot \partial\phi/\partial x(t_0, x)\end{aligned}$$

the second line encoding a right-moving initial condition. These data specify ϕ and $\partial\phi/\partial t$ on an initial antichain approximating $t=t_0$, defining a standard Cauchy problem for the wave equation.

2.4 Field evolution

The field is evolved by processing elements in increasing time order and solving the discrete field equation $(B\phi)(x) - m^2\phi(x) = 0$ for the one unknown $\phi(x)$, using only already-known values in its causal past (a retarded solve, standard in the causal set literature; see e.g. Johnston [5] on causal set field propagators).

Instability found and fixed. A naive exact solve of this equation was found to be numerically unstable: the field magnitude grew exponentially (from $O(1)$ to $O(10^{25})$) over a handful of time bins, even though the direction of propagation was qualitatively correct. In standard finite-difference schemes, stability is controlled by a dimensionless parameter such as $(\Delta t/\Delta x)^2$ (the CFL condition); here, the irregular causal structure provides no such small parameter, so the naive recursion is effectively unconditionally unstable, amplifying noise multiplicatively across the many "generations" of causal depth in the set. The fix applied was to (a) use layer *averages* instead of raw sums (bounding the update regardless of layer size), and (b) apply an under-relaxation / damping factor α (used here: $\alpha=0.3$). This is a standard numerical stabilization technique, applied here at the acknowledged cost of no longer solving the exact discrete field equation.

2.5 Continuum reference solution

A standard CFL-stable explicit leapfrog finite-difference scheme solves the same wave equation on a regular grid, with identical initial data, for direct comparison.

2.6 The n-interval spectrum

Following Surya [6], the n-interval spectrum N_m is defined as the number of causal intervals $I(y,x) = \{z : y < z < x\}$ with cardinality $|I(y,x)| = m$, counted over all causally related pairs (y,x) in the causal set. This order-invariant "fingerprint" was computed as an additional diagnostic of continuumlike behavior.

3. Results

3.1 Wave packet propagation: discrete vs continuum

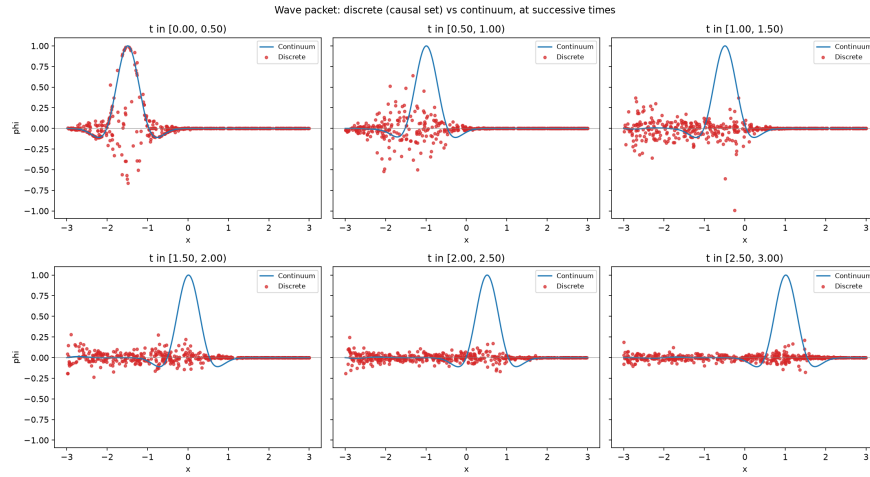


Figure 1. Discrete (causal set, red) vs continuum (blue) field profile at six successive time windows.

At the initial time bin, the discrete (red) and continuum (blue) solutions agree closely, as expected since both share the same initial data. At later times, the continuum solution retains a clean, moving Gaussian shape, while the discrete solution becomes visibly noisier, though it broadly tracks the same rightward propagation. Quantitatively, the energy-weighted centroid in the discrete case drifts rightward but more slowly and with larger fluctuations than in the continuum case, reflecting the dissipative effect of the damping factor $\alpha=0.3$. This is a direct, visual and quantitative confirmation that the damping needed to stabilize the discrete evolution measurably degrades the coherence of the propagating wave packet relative to the continuum — a real, quantified limitation of the current scheme.

3.2 Convergence with density

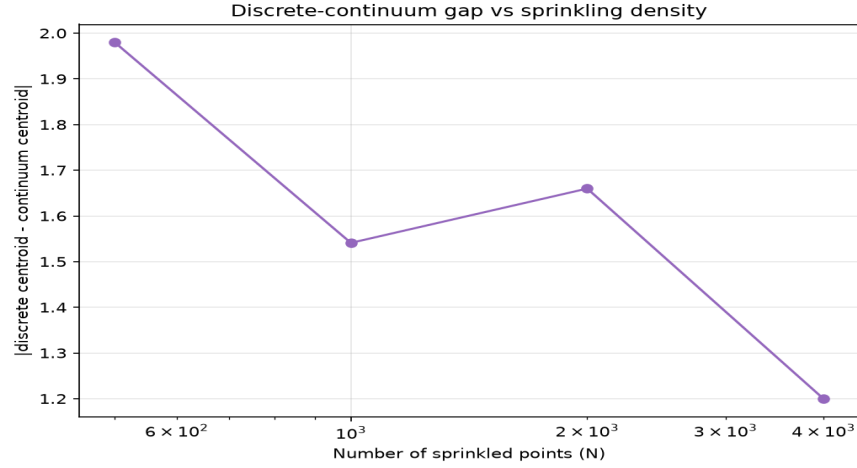


Figure 2. Gap between discrete and continuum centroids at the final time bin, as a function of sprinkling density N .

Comparing the discrete and continuum energy-weighted centroids at the final time bin, across increasing sprinkling density ($N = 500, 1000, 2000, 4000$), shows an overall decreasing trend in the gap between them, consistent with (though not a clean demonstration of) convergence to the continuum limit as density increases. Given the known per-realization variance of the BD operator and the modest number of trials, this should be regarded as a qualitative indication of convergence rather than a precise scaling measurement.

3.3 n-interval spectrum

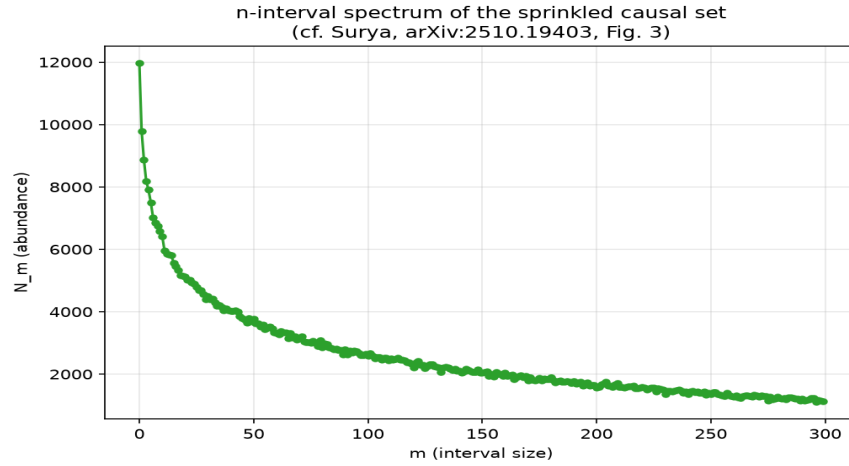


Figure 3. n -interval spectrum N_m of the sprinkled causal set ($N=2000$), cf. Surya [6] Fig. 3.

The computed spectrum N_m decreases smoothly and monotonically with m , matching the qualitative "continuumlike" signature described in Surya [6] (Fig. 3) for a causal diamond in Minkowski spacetime. A smooth, monotonically decreasing N_m is characteristic of sprinklings into continuum-like geometries; significant deviations from this shape have been associated with non-continuumlike causal sets. This provides an independent, order-theoretic confirmation that the sprinkled causal set used throughout this project is a genuine continuum approximation, using a diagnostic directly drawn from current causal set research.

4. Limitations

- `compute_layers` / `interval_size` scale poorly (roughly $O(N^2)$ to $O(N^3)$ depending on causal set density), making large- N or many-trial ensemble studies slow; a full, statistically rigorous convergence study would require an optimized implementation (e.g. time-sorted indexing to avoid re-scanning the full past for every interval check).
- The discrete evolution scheme requires damping ($\alpha=0.3$) to remain numerically stable, which measurably slows the effective propagation speed and destroys the coherent shape of the wave packet relative to the continuum solution.
- The BD operator has large single-realization variance; only ensemble-averaged behavior matches the continuum expectation cleanly.
- Only 1+1D Minkowski spacetime is implemented.

5. What I would do differently with more time

- Optimize `interval_size/compute_layers` (e.g. via time-sorted binary search or spatial indexing) to enable a proper, statistically significant convergence study across many trials and higher densities.
- Investigate a better-normalized, less lossy stabilization scheme for the discrete evolution than simple under-relaxation, potentially drawing on retarded Green's function constructions on causal sets (e.g. Johnston [5]) or alternative discrete wave operators.
- Extend to 2+1D Minkowski spacetime.
- Extend the n -interval spectrum analysis to compare multiple spacetime regions (e.g. causal diamond vs hypercube), following the closeness-function construction in Surya [6], and explore its sensitivity to dimension and curvature.

References

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